REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

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1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

Metric	Value
Total Network Flows	11574
Normal Flows	11016
Detected Attacks	558
Attack Rate	4.82%

2. Machine Learning Model

Parameter	Value
Algorithm	Random Forest Classifier
Training Features	duration, protocol_type, src_bytes, dst_bytes, count
Classification	Normal / Attack
Optimized Threshold	0.50
Validation Accuracy	0.76%
Attack Precision	0.96
Attack Recall	0.60
Attack F1-Score	0.74

3. Network Protocol Distribution

Protocol	Flow Count
TCP	3923
UDP	3182
HTTPS	2809
DNS	923
QUIC_OR_UDP443	662
HTTP	50
DHCP	13
IPSEC	8
OTHER	2
ICMP	1
NTP	1

4. Detected Attack History

Time	Source	Destination	Protocol	Threat Vector	Severity	Confidence
2026-09-06 00:48:54	192.168.1.7:57688	8.8.8.8:443	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-06 00:48:53	192.168.1.7:64056	8.8.8.8:443	QUIC_OR_UDP_443	ANOMALOUS FLOW	HIGH	100.0%
2026-09-06 00:48:52	104.16.9.34:443	192.168.1.7:56577	HTTPS	ANOMALOUS FLOW	HIGH	76.0%
2026-09-06 00:48:52	104.16.9.34:443	192.168.1.7:56575	HTTPS	ANOMALOUS FLOW	HIGH	57.39%
2026-09-06 00:48:52	104.16.9.34:443	192.168.1.7:56576	HTTPS	ANOMALOUS FLOW	HIGH	68.0%
2026-08-29 20:33:34	10.94.204.202:60791	10.94.204.69:53	DNS	ANOMALOUS FLOW	HIGH	92.0%
2026-08-29 20:33:34	10.94.204.202:51449	10.94.204.69:53	DNS	ANOMALOUS FLOW	HIGH	88.0%
2026-08-29 20:33:34	10.94.204.202:52013	10.94.204.69:53	DNS	ANOMALOUS FLOW	HIGH	74.0%
2026-08-29 20:33:34	10.94.204.202:54332	10.94.204.69:53	DNS	ANOMALOUS FLOW	HIGH	70.0%
2026-08-29 20:33:33	142.251.151.119:443	10.94.204.202:55747	QUIC_OR_UDP_443	ANOMALOUS FLOW	HIGH	73.59%
2026-08-29 19:53:23	10.94.204.202:65368	172.217.118.4:443	HTTPS	ANOMALOUS FLOW	HIGH	77.36%
2026-08-29 19:53:04	10.94.204.202:55082	140.82.113.21:443	HTTPS	ANOMALOUS FLOW	HIGH	50.34%
2026-08-29 19:48:54	10.94.204.202:65368	172.217.118.4:443	HTTPS	ANOMALOUS FLOW	HIGH	73.89%
2026-08-29 19:40:42	10.94.204.69:53	10.94.204.202:61152	DNS	ANOMALOUS FLOW	HIGH	84.67%
2026-08-29 19:39:49	10.94.204.69:53	10.94.204.202:53580	DNS	ANOMALOUS FLOW	HIGH	53.86%
2026-08-29 19:38:55	10.94.204.202:62659	172.64.155.209:443	HTTPS	ANOMALOUS FLOW	HIGH	53.38%
2026-08-29 18:17:22	10.94.204.202:60479	172.64.155.209:443	HTTPS	ANOMALOUS FLOW	HIGH	54.24%
2026-08-26 22:10:09	192.168.1.1:53	192.168.1.5:52851	DNS	ANOMALOUS FLOW	HIGH	93.11%
2026-08-26 22:10:09	192.168.1.1:53	192.168.1.5:60211	DNS	ANOMALOUS FLOW	HIGH	93.11%
2026-08-26 22:10:09	192.168.1.1:53	192.168.1.5:63849	DNS	ANOMALOUS FLOW	HIGH	85.45%

5. Security Recommendations

1. Investigate repeated attack predictions from the same source address.

2. Monitor network flows with unusually high traffic volume or packet counts.

3. Review suspicious TCP and UDP connections identified by the ML classifier.

4. Use the attack confidence score together with network context when validating alerts.

5. Continue collecting representative network traffic to improve the IDS model.

6. Periodically retrain and evaluate the model using updated datasets.

6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.